
When the World Changes but the Score Does Not: A Counterfactual Audit of Computer-Use Agents

Vinh Duc Tran

Hanoi University of Science and Technology
tranducvinh2004@gmail.com

Abstract

Computer-use agents are typically scored by whether an episode finishes and what a rubric judge assigns. That scalar mixes three questions: termination, whether the final answer reflected the world the agent saw, and whether the judge noticed when that world changed. We separate them with paired counterfactual episodes on MyPCBench: a locked intervention rewrites a pre-registered determining set D while instruction, interface, and judge stay fixed. Tracking is read from the final answer against guest gold, never from the score; score sensitivity is exact equality $S_{CF} = S_{Base}$ versus any nonzero delta. Across agents, successful counterfactual tracking sometimes leaves the benchmark score unchanged and sometimes changes it; conversely, unchanged high scores can occur despite incomplete state tracking. On a frozen 5×10 design (50 intended cells; 4 never scheduled; 46 executed; 92 legs; 35 execution failures; 24 valid pairs), Claude has 9 valid pairs (7 tracking-valid: 6 Type A, 1 score-sensitive, 2 Type B; invariance $6/7 = 0.857$, Clopper-Pearson 95% CI $[0.421, 0.996]$); GPT-5.5 has 8 valid (7 tracking-valid: 3 Type A, 4 score-sensitive, 1 Type B; $3/7 = 0.429$, $[0.099, 0.816]$); Qwen3.5-35B-A3B has one valid pair (Type A; $[0.025, 1.000]$). Qwen3.5-9B (ablation) and Qwen3.8-Flash (exploratory) are not pooled into these rates. A stable score does not certify tracking; a moving score does not certify a tracking failure.

1 Introduction

Leaderboard evaluation of computer-use agents (CUAs) reports a completion score: the episode ended, and a rubric assigned a number [Jang et al., 2026, Xie et al., 2024]. That number is asked to stand in for three facts: whether the harness recorded a terminal DONE; whether the answer matched the environment that was present; and whether the judge’s score moved when that state moved.

These facts need not agree. An agent can finish, miss a manipulated field, and still score 100 if the rubric does not pin that field. An agent can finish, report the field correctly on both sides of a paired comparison, and still see the score swing for reasons orthogonal to that field. Inferring tracking or judge sensitivity from the scalar alone is therefore undefined.

This paper makes the three facts separately observable. For each frozen task we run a base episode in E^0 and a counterfactual episode in $E^1 = I(E^0)$, where I is a locked SQL (and, when designed, file) patch on a pre-registered determining set D . Prompt, UI, screenshot size, and judge are held fixed. Guest probes confirm that gold moved. We code *completion* from the last trajectory action, *tracking* from the final answer against guest gold, and *score sensitivity* from exact scalar equality. Incomplete runs are execution failures, not tracking misses.

Contributions. (1) An audit taxonomy of valid pairs: Type A (tracked, score-invariant), score-sensitive (tracked, score moves), and high-score tracking failure (Type B: incomplete tracking with a high, invariant score). (2) Empirical separation on a frozen ten-task MyPCBench audit across

Table 1: Taxonomy of valid pairs. Rows: agent vs. gold. Columns: exact S . Execution failures are excluded.

	Score invariant	Score moves
Tracks D	Type A	Score-sensitive
Misses D	Type B (high, invariant)	not a headline class here

three primary lanes, with a size ablation and an exploratory lane held out of the primary rates. (3) Pre-registered intervention roles predicted what state was manipulated, not how the agent would respond to that manipulation. The claim is methodological and local: we do not estimate a population failure rate and we do not headline a pooled invariance fraction.

2 Setup

Intervention. For task t , E_t^0 is the seeded guest and $E_t^1 = I_t(E_t^0)$ is the same guest after a locked patch on D_t . Some tasks are dual-channel (sqlite and a paired local file must move together); others hold one channel constant as a distractor. The same agent A , instruction T_t , and judge J produce $\tau^0 = A(T_t, E_t^0)$ and $\tau^1 = A(T_t, E_t^1)$. Every intervention in this snapshot has a confirming guest probe.

Reads. $G(\tau)$ is the task-relevant quantities in the final answer. $S(\tau) = J(\tau) \in \{0, \dots, 100\}$. Writer-side track fields, `result.txt`, and the mere presence of a score file are not used for DONE, tracking, or validity. `done`(τ) = 1 iff the last recorded action is literally DONE. A pair is valid iff both legs are done. `track` = 1 iff $G(\tau^0)$ matches D_t on E_t^0 and $G(\tau^1)$ matches D_t on E_t^1 ; joint D_t requires every component on both legs. Tracking is never inferred from S . On a tracked pair, invariance is $S(\tau^1) = S(\tau^0)$ with no tolerance.

Classes. These labels are an audit taxonomy for this protocol, not a claim that CUAs have three natural kinds. Type A: `track` = 1 and $\Delta S = 0$. Score-sensitive: `track` = 1 and $\Delta S \neq 0$. Type B (high-score tracking failure): `track` = 0 with a high, invariant score. The high-score cutoff used here is operational: both legs ≥ 80 and $\Delta S = 0$ on the three primary Type B cells (100→100 on the ablation/exploratory Type B cells). A different cutoff would relabel the same trajectories; we report the scores, not a law of nature. Type A is not automatically a rubric bug; Section 3.6 reads per-criterion arrays where they exist.

Universe. Ten tasks from one pre-registered eligibility pool (Table 2). Designed roles were frozen before runs. Lanes: Claude, GPT-5.5, Qwen3.5-35B-A3B (primary); Qwen3.5-9B (ablation); Qwen3.8-Flash (exploratory). Design matrix: 50 intended cells. Never scheduled (not execution failures): Qwen3.5-9B and Qwen3.8-Flash on `preference_inference-f018` and `counterfactual-f004` (4 cells). Executed: 46 cells, 92 legs; 57 DONE, 35 execution failures (15 of 35 on Qwen3.5-35B-A3B, mostly empty tool XML). Valid pairs: 24. Invariance rate = Type A / tracking-valid. Intervals are exact Clopper–Pearson 95% CIs. Lanes are never pooled.

Compute. This paper does not train models and performs no additional agent runs. Claude and GPT-5.5 cells used provider computer-use APIs; Qwen cells used a hosted API (OpenRouter). Original wall-clock, GPU type, and cluster identifiers were not instrumented for this audit.

3 Results

3.1 Type A: tracked, score unchanged

All five lanes complete `retrieval-f001` (loyalty tier/miles Gold/38,450 → Silver/8,620) and track both legs; scores stay at the base value (100 on four lanes; 80 on Qwen3.5-9B). The rubric does not pin Gold or 38,450.

Claude `aggregation-f018` was designed as a Type B trap (sqlite charitable \$950 → \$0 and home-office 104 → 0 days; W-2/1099 and files held). Claude is the only lane that completed both legs.

Table 2: Frozen tasks and pre-registered design role.

Task	Phase	Designed role
retrieval-f001	A	control
aggregation-f003	A	control
preference_inference-f018	A	designed Type B (joint D)
counterfactual-f004	A	score-sensitive contrast
retrieval-f003	B	control
retrieval-f016	B	control
retrieval-f029	B	control (dual-channel)
retrieval-f030	B	designed Type B
aggregation-f018	B	designed Type B
preference_inference-f004	B	score-sensitive contrast

Observed class is Type A: CF zeroes both sqlite fields, holds the untouched file, and flags the mismatch; 100 $\rightarrow$ 100. This is the cleanest positive control.

Other primary Type A pairs: Claude retrieval-f003 65 $\rightarrow$ 65, retrieval-f016 100 $\rightarrow$ 100, retrieval-f029 100 $\rightarrow$ 100, aggregation-f003 80 $\rightarrow$ 80; GPT retrieval-f001, retrieval-f003 100 $\rightarrow$ 100, aggregation-f003 50 $\rightarrow$ 50; Qwen3.5-35B-A3B retrieval-f001 100 $\rightarrow$ 100.

3.2 Score-sensitive: tracked, score moves

preference_inference-f004 flips HangryDash from Cooper’s Seafood House (88) to Backyard Ale House (59), TableFind held. Claude and GPT both report the flip. Scores drop 100 $\rightarrow$ 79 and 100 $\rightarrow$ 58 because the rubric pins Cooper’s as HD top.

GPT also tracks three retrieval interventions whose scores move: retrieval-f016 100 $\rightarrow$ 85; retrieval-f029 33 $\rightarrow$ 100; retrieval-f030 53 $\rightarrow$ 100 (TY2025 1099 held at \$1,200; charitable \$950 $\rightarrow$ \$100). Type A and score-sensitive co-exist for the same agent and protocol.

3.3 Type B: high-score tracking failure

1. Claude retrieval-f030, 100 $\rightarrow$ 100. Base answers filed TY2024 1099-NEC \$1,080, not determining TY2025 \$1,200. CF reports \$1,200 and charitable \$100. The miss is the base-leg year, not CF charitable.
2. Claude preference_inference-f018, 100 $\rightarrow$ 100. Joint D : GME 85 $\rightarrow$ 0 and OddsMarket YES 200/active $\rightarrow$ 0/settled. CF lists GME 0 and still lists GameStop YES as 200 / \$16.00.
3. GPT counterfactual-f004, 87 $\rightarrow$ 87. Guest zeroes the 1099; CF still reports a \$1,200 stipend and \$1,068/year net.

Ablation/exploratory Type B (not primary): Qwen3.5-9B retrieval-f016 base \$8,788.75 vs. gold \$8,213.25 (CF exact), 100 $\rightarrow$ 100; Qwen3.8-Flash retrieval-f029 CF “Total Income \$91,200” vs. Box 1 \$90,000, 100 $\rightarrow$ 100.

3.4 Designed role vs. observed class

Pre-registered intervention roles predicted what state was manipulated, not how the agent would respond to that manipulation. aggregation-f018 (designed Type B) $\rightarrow$ Claude Type A. retrieval-f030 (designed Type B) $\rightarrow$ Claude Type B and GPT score-sensitive. preference_inference-f018 $\rightarrow$ Claude Type B as specified. preference_inference-f004 $\rightarrow$ Claude and GPT sensitive. counterfactual-f004 (designed sensitive) $\rightarrow$ GPT Type B.

3.5 Per-model counts

Qwen3.5-35B-A3B’s 1/1 is execution coverage (nine of ten cells fail), not an invariance finding.

Table 3: Primary tier. Invariance = Type A / tracking-valid. Exact Clopper–Pearson 95% CI. $n \leq 9$.

Model	Valid	Tracking-valid	Type A	Sensitive	Type B	Invariance	95% CI
Claude	9	7	6	1	2	0.857	[0.421, 0.996]
GPT	8	7	3	4	1	0.429	[0.099, 0.816]
Qwen3.5-35B-A3B	1	1	1	0	0	1.000	[0.025, 1.000]

Table 4: Ablation and exploratory. Not pooled with Table 3.

Model	Valid	Track.	A	Sens.	B	Inv.	95% CI
Qwen3.5-9B	3	2	1	1	1	0.500	[0.013, 0.987]
Qwen3.8-Flash	3	2	1	1	1	0.500	[0.013, 0.987]

3.6 Judge vs. agent

On aggregation-f003 (\$4,871.70 $\rightarrow$ \$400), four completing lanes track both totals. Claude and GPT are Type A; 9B and Flash are score-sensitive. GPT stays at [1, 1, 0, 0] both legs: refund-linked criteria pass twice; year-count criteria never fire. 9B and Flash each pick up one extra criterion under CF. Type A is not “the refund criteria were blind”; score-sensitive is not “the agent failed to track.”

4 Discussion

Completion scores measure termination plus whatever the rubric binds. They do not measure whether D was read, and they do not measure whether J is attached to D . Type A shows tracking without score movement; score-sensitive shows both; Type B is a high-score tracking failure under this taxonomy, not a fundamental category of CUA behavior.

Score invariance is not a defect to patch out of every rubric. A channel-invariant item that asks whether the agent used the right app and reported a value can award the same score to two correct worlds (retrieval-f001). The same agent can be Type A on some tasks and score-sensitive on others (GPT on retrieval-f001/retrieval-f003/aggregation-f003 vs. retrieval-f016/retrieval-f029/retrieval-f030/preference_inference-f004), so the protocol does not merely detect a globally blind judge.

Broader impacts. Positive: the audit makes hidden evaluation failure modes observable and encourages treating completion scores as insufficient certificates of state tracking. Negative: the same pairing could be used to probe proprietary graders; we intend the results as a diagnostic of scoring, not as an adversarial exploit of any deployed system.

5 Related work

OSWorld [Xie et al., 2024], WebArena [Zhou et al., 2024], and VisualWebArena [Koh et al., 2024] score agents with completion rubrics. MyPCBench [Jang et al., 2026] seeds a personal desktop that the interventions reuse. OSWorld is Apache-2.0; WebArena is Apache-2.0; VisualWebArena is MIT; MyPCBench is MIT with an OSWorld-derived harness under Apache-2.0. None of these leaderboards pair E^0 with $E^1 = I(E^0)$ to split tracking from S . Shao et al. [2026] audit whether intended capability remains necessary for a score; we audit whether a gold-moving intervention appears in the answer and in S . Turk [2026] find that coverage-style scores can hide update failures under controlled mutations; we observe the same pattern for GUI CUAs. Bellibatlu et al. [2026] measure judge instability under prompt paraphrase; we measure sensitivity to a verified world-state change. The pairing follows interventionist language [Pearl, 2009]; we do not identify model internals.

6 Limitations

Ten tasks, one environment, five lanes, $n \leq 9$ valid pairs per primary model. CIs bound binomial uncertainty on this sample, not a deployment rate. Qwen3.5-35B-A3B contributes one semantic pair. All 24 valid pairs were hand-coded from final-answer text against guest gold. Score deltas are

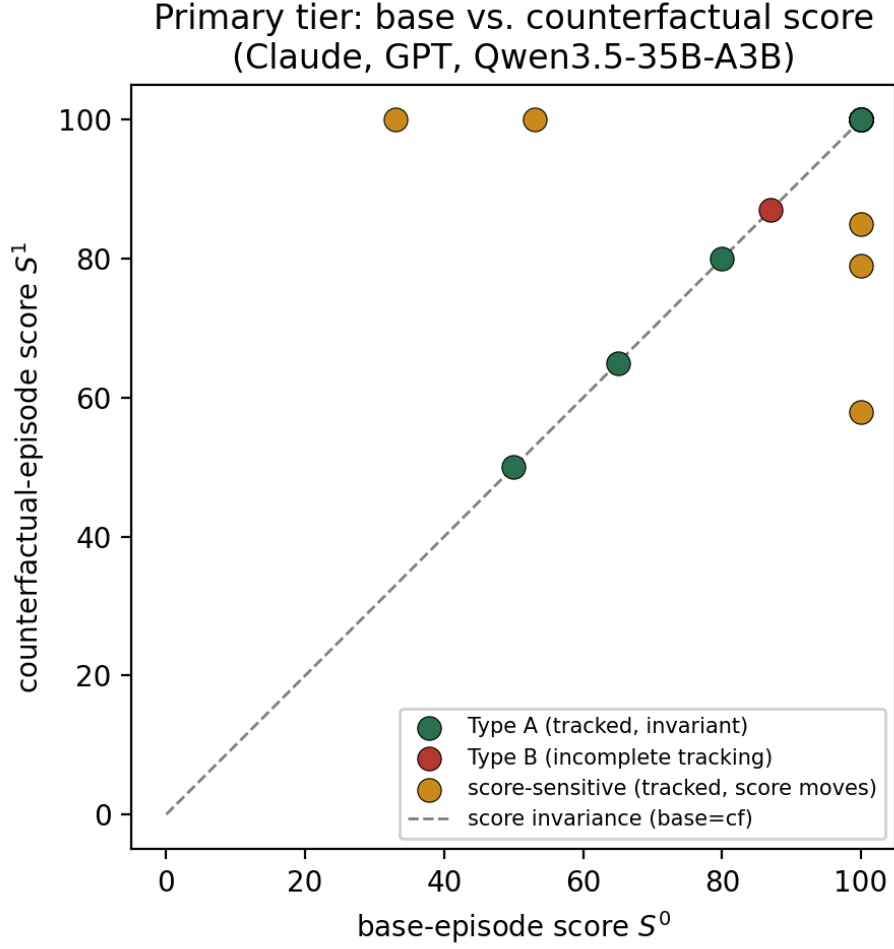


Figure 1: Primary valid pairs: S^0 vs. S^1 . Diagonal: score invariance. Off-diagonal tracked pairs are score-sensitive; high, invariant misses of D are Type B.

attributed to rubric items only where per-criterion arrays exist. Interventions test controlled state changes, not arbitrary real-world shifts.

7 Conclusion

On this frozen audit, task completion, counterfactual tracking, and benchmark score sensitivity come apart. A stable score does not certify tracking; a moving score does not certify a tracking failure. Paired gold-moving checks make that split observable; CUA leaderboards currently do not run them.

References

- Rohith Reddy Bellibatlu, Edward Raff, and Wenbin Zhang. JudgeSense: A benchmark for prompt sensitivity in LLM-as-a-judge systems, 2026. URL <https://arxiv.org/abs/2604.23478>.
- Lawrence Keunho Jang, Andrew Keunwoo Jang, Jing Yu Koh, and Ruslan Salakhutdinov. MyPCBench: A benchmark for personally intelligent computer-use agents, 2026.
- Jing Yu Koh, Robert Lo, Lawrence Jang, Vishakh Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. VisualWebArena: Evaluating

- multimodal agents on realistic visual web tasks. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.
- Jiaqi Shao, Hanck Chen, Wei Zhang, Maxm Pan, and Bing Luo. Do agent benchmarks measure capability? protocol validity in the age of agentic AI, 2026. URL <https://arxiv.org/abs/2607.22368>.
- Matt Turk. Counterfactual evaluation reveals hidden capability profiles in clinical LLMs and agents, 2026. URL <https://arxiv.org/abs/2605.30590>.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Daoguang Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. In *Advances in Neural Information Processing Systems*, 2024.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*, 2024.